

- **Composite resin Restoration**



outline

- Definition
- Classification of composite
- Types
- Recent advances in composites
- Indication and Contraindication
- Steps in composite restoration
- Failures of composite filling
- Sandwich technique

- **Definition:** Composite restorations are tooth-colored dental fillings made from a mixture of plastic resins and glass particles.
- **Purpose:** Primarily used to restore decayed or damaged teeth
- **Advantages:** Aesthetic, conservative, minimal tooth removal, and easy shaping.

classification

- Based on curing mechanism
- Based on particle size
- Based on viscosity
- Based on Application

- Chemically activated
- Light activated
- Dual cure

- Macrofillers (10-100 μm)
- Midifillers (1-10 μm)
- Minifillers (0.1-1 μm)
- Microfillers (0.01-0.1 μm)
- Nanofillers (0.005-0.01 μm)
- Hybrid (range of sizes v
- Flowable
- Packable

- Restorative composites-

- Hybrids (*FI* [redacted])

- Microfilled (*FI* [redacted])

- Flowable (*FI* [redacted])

- Packable

- Nanocomposites (*FI* [redacted])

- Core build-up composite

- Prosthodontic composite

- Provisional composites

- Luting composites

Types of Composite Materials

1. Macrofilled Composites

- Larger filler particles.
- Less polished and more prone to wear.
- Used in posterior teeth.

2. Microfilled Composites

- Smaller filler particles.
- Excellent polish and aesthetic appeal.
- Used in anterior teeth.

3. Hybrid Composites

- Combination of micro and macro filler particles.
- Balanced strength and aesthetic properties.

4. Nanocomposites

- Nanoparticles for superior polish and strength.
- Used for both anterior and posterior restorations.

RECENT ADVANCES IN COMPOSITES

Composites

- **Flowable composite resins**
- **Condensable (packable) composites**
- **Antibacterial/ion releasing composites**
- **“Smart” composites**
- **Giomers**

Flowable Composite Resins

- They were introduced in the dentistry in **late 1996**.
- These resins have **lower filler content** ranging from **41-53% by volume**.
- Low filler loading is responsible for **decreased viscosity of composites**, which allows them to be injected into both preparation, this makes them a good choice as a pit and fissure restorative.
- They have an **average particle size** ranging from 0.04-1 μm .
- Their decreased viscosity is achieved by reducing the filler volume.

Indications

- Preventive resin restorations
- Small pit and fissure sealants
- Small, angular class V lesions
- For repairing ditched amalgam margins
- Repair of small porcelain fractures
- Inner layer for class II posterior composite resin placement for sealing the gingival margin
- Resurfacing of worn composite or glass ionomer cement restorations

- **For repair of enamel defects**
- **For repair of crown margins**
- **Repair of composite resin margins**
- **For luting porcelain and composite resin veneers**
- **Class I restorations**
- **Small class III restorations**
- **As base or liner**
- **Tunnel restorations.**

Advantages

- **Low viscosity**
- **Improve marginal adaptation** of posterior composites by acting as an elastic, stress absorbing layer over which composite is placed
- **High wettability** of the tooth surface
- **High depth of cure**
- **Penetration** into every irregularity of preparation
- **Ability to form layers of minimum thickness**, thus eliminate air entrapment
- **High flexibility**, so less likely to be displaced in stress concentration areas

Cont.....

- Radiopaque
- Availability in **different colors**
- Require **minimally invasive tooth preparations**
- Used in narrow and deep pits and fissures (For the pediatric patient).

Disadvantages

- More susceptible to wear in stress-bearing areas
- Weaker mechanical properties
- More polymerization shrinkage and wear
- Sticks to the instrument, so difficult to smoothen the surface.

Condensable (Packable) Composites

- Developed in an attempt to **improve the compressive, tensile and edge strength and handling of the composite.**
- Have high viscosity and used for posterior tooth preparation and have greater control over the proximal contour of class II preparations.
- Filler content ranges from 48-65% by volume.
- Average particle size ranges from 0.7-20 μm .
- Possess **improved mechanical properties** because of presence of ceramic fibers.
- It's consistency is like freshly triturated amalgam. Each increment can be condensed similar to amalgam restoration and can be cured to a depth of over 4 mm.

Indications

- Indicated for stress-bearing areas
- In class II restorations as they allow easier establishment of physiological contact points.

Advantages

- Increased wear resistance because of presence of ceramic fibers
- Condensability like silver amalgam restoration
- Greater ease in achieving a good contact point
- Produce better reproduction of occlusal anatomy
- Deeper depth of cure because of light conducting properties of individual ceramic fibers
- High flexural modulus
- Decreased polymerization shrinkage because of presence of ceramic fibers
- Reduced stickiness
- Physical and mechanical performance is similar to that of silver amalgam.

Disadvantages

- Difficulties in adaptation between one composite layer and another.
- Poor aesthetics in anterior teeth.

Giomers

- **is hybrid of words** “glass ionomers” and “composite”. These are relatively new type of restorative materials.
- They are also known as PRG composites (Prereacted glass ionomer composites).
- They have properties of both glass ionomers (fluoride release, fluoride recharge) and resin composite (excellent aesthetics, easy polishability, biocompatibility).

Indications

1. Noncarious cervical lesions
2. Root caries
3. Deciduous tooth caries.

Properties of Giomers

1. Easier to polish than glass ionomers
2. Optimum fluoride release
3. Excellent aesthetics
4. Better surface finish
5. Chemical bonding to tooth structure
6. Biocompatibility
7. Sensitive to moisture and desiccation.

Compomers

- **Polyacid Modified Composite Resins.**
- Compomers are class of dental materials that provide combined advantages of composites (term 'Comp' in their name) and glass ionomer ('Omers' in their name).

Composition

- **Resin matrix:** Dimethacrylate monomers with two carboxylic groups present in their structure
- **Filler:** Reactive silicate glass containing filler
- **Photo initiators and stabilizers**
- There is no water in the composition and ion leachable glass is partially silanized to ensure bonding to matrix.

Properties of compomer

- Their characteristics are very similar to composite resins.

Adhesion:

- Adhesion to tooth structure is by micromechanical means and requires acid etching and use primer/adhesive.

Physical properties:

- Physical properties such as strength, fracture toughness are very much similar to composites.

Bond strength: It is similar to composite.

Adaptation: at cervical margin is similar to composite resins.

Fluoride release:

- It is greater than composite resins but less than glass ionomer systems.
- They initially release high levels of fluorides but after some time the level falls rapidly to low level.

Color matching and optical properties: are superior to glass ionomer cements.

Advantages

- Optimal aesthetics
- Easy to handle
- Easy to polishing
- Easy to place
- Require no mixing
- Bond strength is higher than glass ionomers.

Disadvantages

- Require use of bonding agent
- Technique sensitive
- Limited fluoride release
- Microleakage more than resin modified glass ionomers
- Expansion of matrix due to water absorption
- Physical properties decrease with time.

ORMOCER

- It is organically Modified Ceramic.
- It is an **organically modified non metallic inorganic composite material.**
- It is three dimensionally cross-linked copolymer.
- First time, it was introduced as dental restorative material in 1998.

Composition of ORMOCER

- Have both organic and inorganic networks.
- They are characterized by presence of three main units:
 - ✓ Organic molecules segment having methacrylate groups which form a highly cross-linked matrix.
 - ✓ Inorganic condensing molecules to make three dimensional network which is formed by inorganic poly-condensation.
 - ✓ fillers

Indications :

1. Restoration for all types of preparations.
2. For esthetic veneers.
3. As orthodontic bonding adhesive.

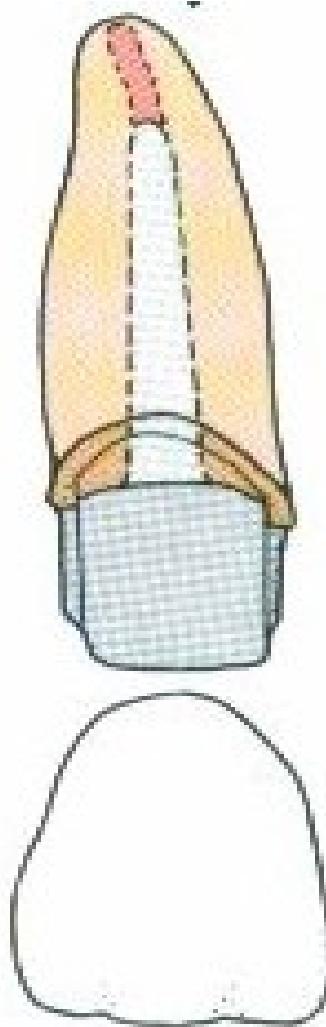
Properties of ORMOCER:

1. More biocompatible than conventional composites.
2. Higher bond strength.
3. Polymerization shrinkage is least among resin-based filling material.
4. Highly esthetic, comparable to natural tooth.
5. High compressive (410 MPa) and transverse strength (143 MPa).

Indications for Using Composite Resins

1. Class-I, II, III, IV, V & VI restorations.
2. Foundations or core buildups.
3. Pit and fissure Sealant & Preventive resin restorations.
4. Esthetic enhancement procedures.
 - Veneering -Partial or Full veneers
 - Tooth contour modifications
 - Diastema closures
5. Luting /Cementation (for indirect restorations)
6. Periodontal splinting.
7. Orthodontic bonding

FOUNDATION OR CORE BUILDUP



Veneer placed to reduce discoloration and to cover the stain



Veneers placed to close diastema



Cont.

- Composite restoration of fractured anterior tooth



Cont.

- Composite used for periodontal splinting of mobile teeth



Cont.

- Composite in fixed orthodontic appliances



CONTRAINDICATIONS OF COMPOSITES

1. When isolation of operating field is difficult.
2. Where very high occlusal forces are present.
3. Class V lesions- where aesthetics is not the primary concern.
4. Poor technical skill- of operator for the restoration.
5. Extent Of lesion- when extends up to the root surface.
6. Patients with high caries susceptibility.
7. When preparation extends sub-gingivally.
8. Patients with poor oral hygiene.

Steps of clinical proceduring for composite restoration

- ➔ Local anesthesia
- ➔ Preparation of operating site
- ➔ Composite selection
- ➔ Shade selection
- ➔ Isolation
- ➔ Tooth preparation
- ➔ Bonding
- ➔ Composite placement
- ➔ Polymerization of composite resins
- ➔ Final contouring, finishing and polishing of composite restoration

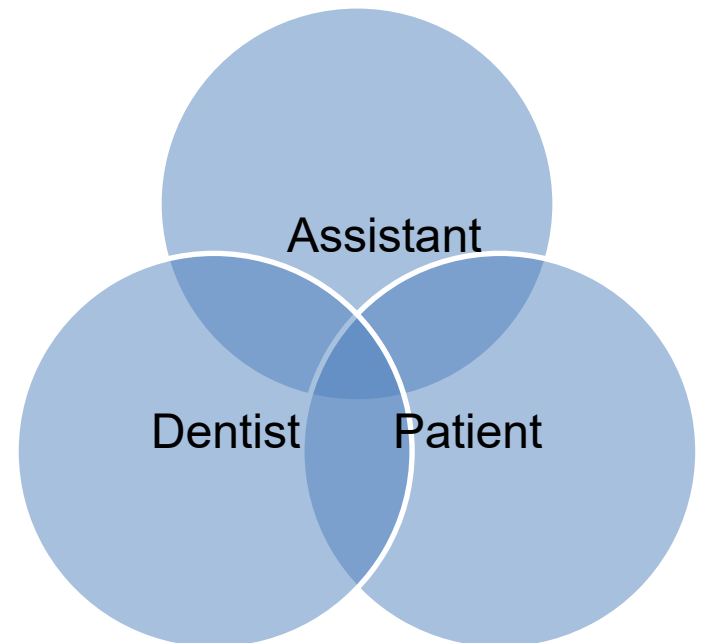
1. Local anesthesia

- Local anesthesia contributes to a pleasant, time saving and uninterrupted procedure.
- patient is more relaxed
- Local anesthesia also helps in marked reduction in salivation
- Decrease gingival bleeding

2.Preparation of operating site

- Clean the operating site with pumice to remove plaque, pellicle, calculus and superficial stains.
- These steps create a site more receptive to etching and bonding

3. shade selection



4. Isolation of operating site

- Isolation for tooth-colored restorations can be accomplished with a rubber dam or cotton rolls, and a retraction cord.
- Contamination of etched enamel or dentin by saliva results in a significantly decreased etching and bonding process.
- Likewise, contamination of the composite material during insertion results in degradation of physical properties.

rubber dam



5. Tooth preparation

- **General concepts for tooth preparation for composite restorations**
 1. Conservation of tooth structure: Tooth preparation is limited to extent of the defect.
 2. Variable depth of pulpal and axial wall : Pulpal and axial walls need not to be flat.
 3. Preparation of operating site: To facilitate bonding, tooth surface is made **rough** by using diamond abrasives.
 4. Enamel bevel: is given in some cases to increase the **surface area** for etching and bonding.
 5. Butt joint on root surface: Cavo surface present on root surfaces has to be butt joint(creating 90 degree).

Cavity preparation

❖ The following three types of design or their combination of cavities are mostly prepared for composite.

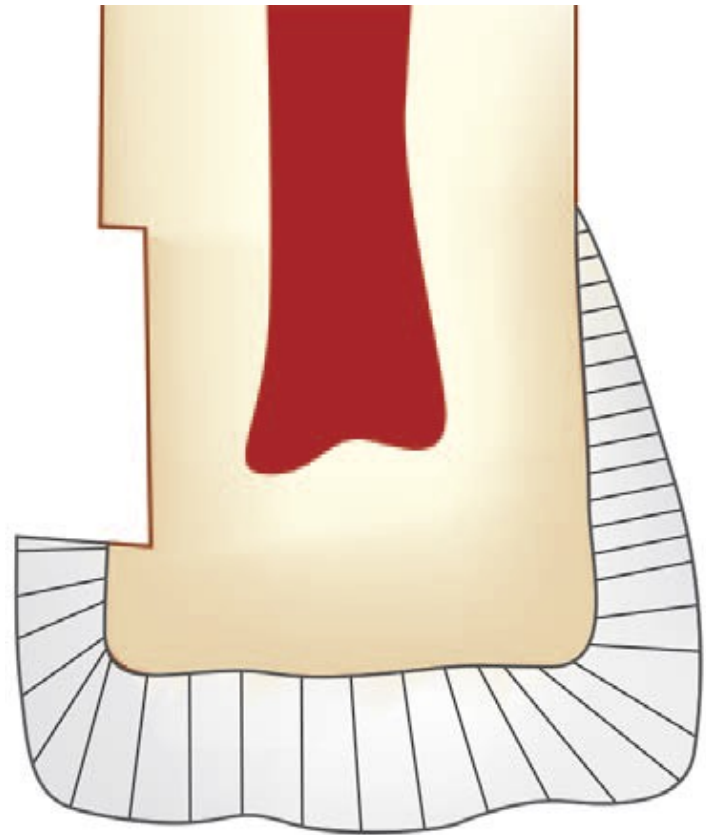
1. Conventional
2. Beveled Conventional
3. Modified (conservative)

A. Conventional

- ❖ Conventional tooth preparations are similar to the cavity preparation to those typical for amalgam restorations.
- A uniform depth of the cavity with 90° cavosurface margin is required
- Except, that there is less outline extension and in tooth preparation walls being rough.

❖ Indication

- Preparations located on root surfaces
- Moderate to large Class I or Class II restorations.

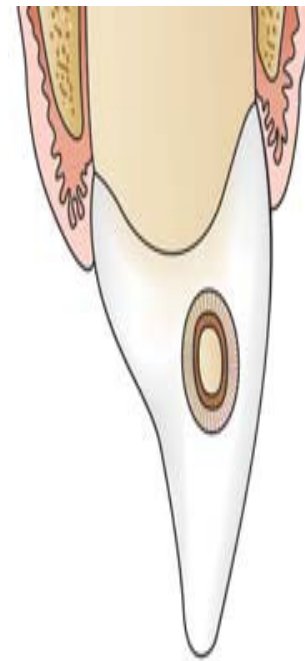


B. Beveled conventional

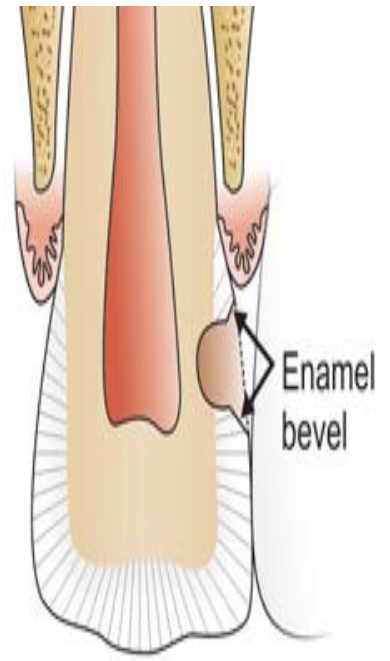
- Beveled conventional tooth preparations are similar to conventional preparations, but with some **beveled enamel margins**.
- Beveling increases favorable end-on etching of enamel prisms and increases the surface area for bonding.

❖ Indication

- to replace an existing restoration exhibiting a conventional design.
- To restore a large preparation
- ✓ especially indicated for classes III, IV and V restorations



A Proximal view



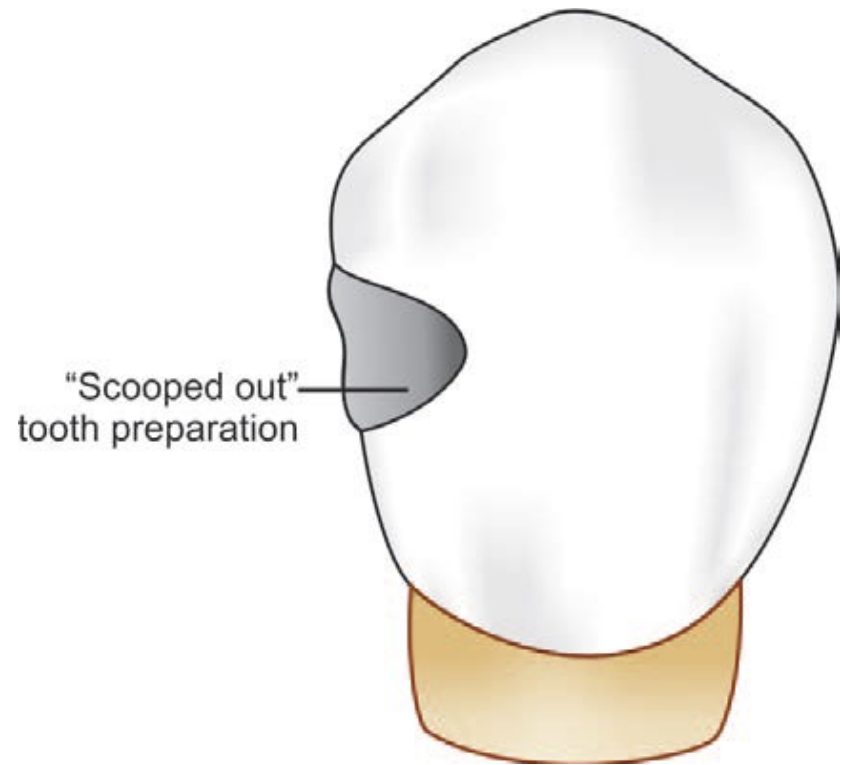
B Labial view

C. Conservative (modified) cavity preparation

- ❖ It is more conservative in nature since retention is achieved by micromechanical bonding to the tooth
- ❖ **Features**
 - ✓ Preparation has scooped out appearance.
 - ✓ It does not have specified wall configuration or pulpal and axial wall depth.
 - ✓ Extent and depth of the preparation depends upon the extent and the depth of carious lesion

❖ Indication

- For initial or small carious lesions



6. Etching and application of bonding

❖ Effects of acid etching-

- Dissolution of interprismatic/core enamel.
- Increase in surface area up to 2000 times.
- Removes the smear layer which increases the wettability and surface energy.
- Removes 10 micrometer of surface enamel and create a microporous layer

❖ **liquid , gel and semi gel** etchants are available, most in concentrations of (30 % - 40%) phosphoric acid

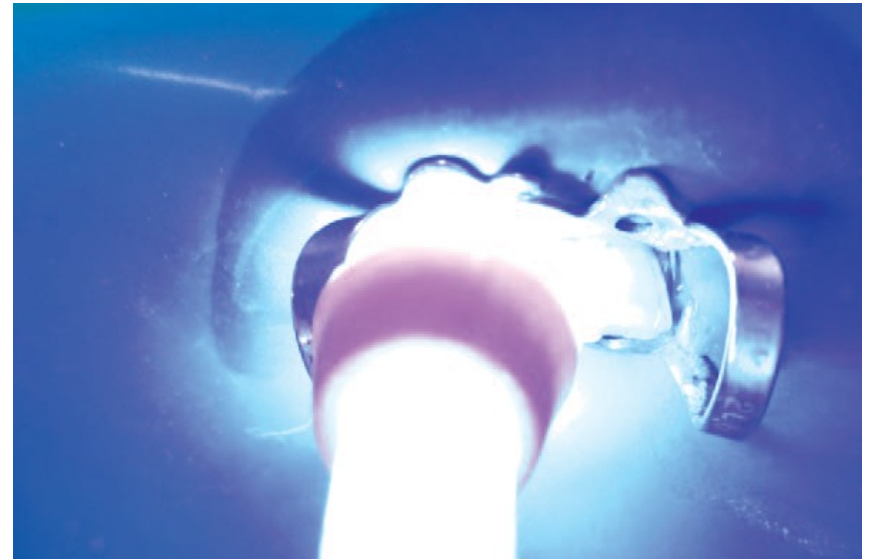
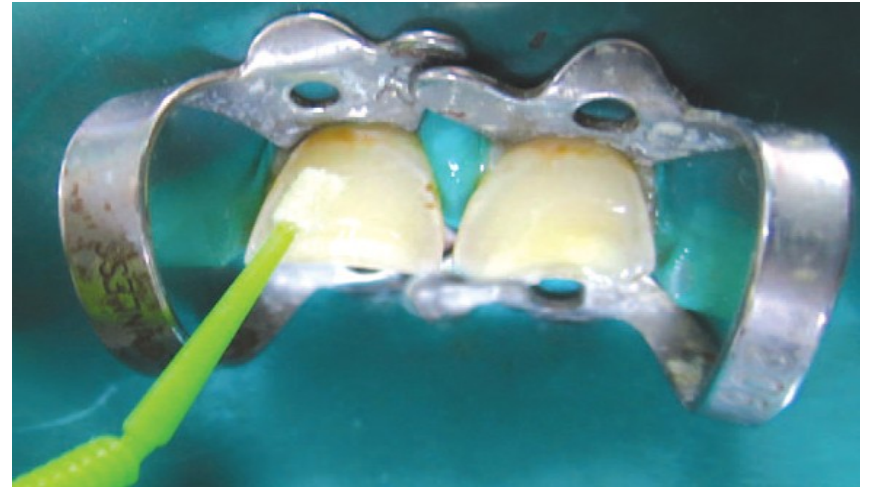
❖ Etching time

- Dentin and enamel-15 sec
- Enamel-30 sec



Bonding agent

- In dentistry, bonding refers to the process of attaching a restorative material to tooth structure by adhesion
- Bonding Agents form a strong bond with the tooth structure and also react with the tooth structure and with the restorative resin.



7. Matrix placement

- The matrix should provide the appropriate proximal contact and contour and prevent major excess of the restorative material beyond the preparation margins on the proximal surface.



8. Inserting composite

- building composites (incremental layers)
- Hand instrument, gun and syringe are used for inserting composite resins



9. curing

- Distance b/n composite and LC1mm
- LC....perpendicular to composite



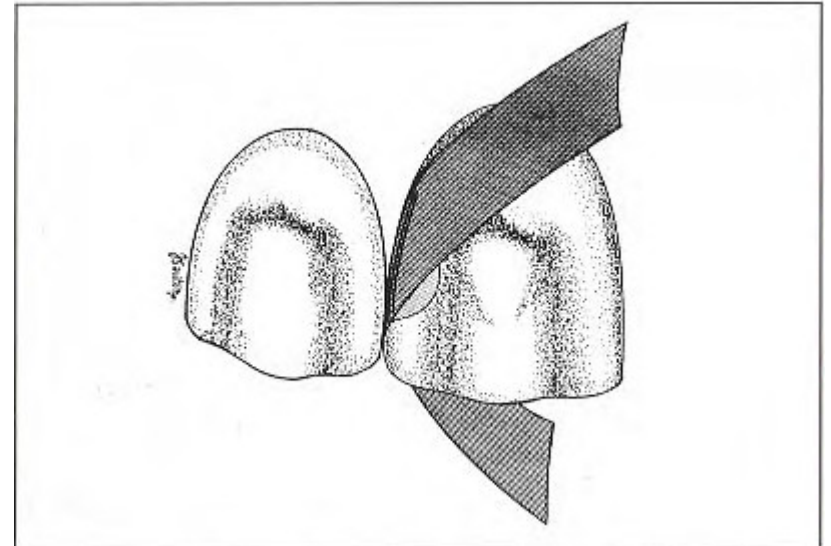
10. Contouring composite

- Good technique and experience in inserting a composite restoration significantly reduces the amount of finishing required.
- Usually a slight excess of material is present that must be removed to provide the final contour and smooth finish.
- Contouring the composite restoration requires **skill** and **knowledge** of correct dental anatomy.

11. Polishing composite

- Polishing the contoured composite restoration is done with **very fine polishing discs, fine rubber points or cups, and/or composite polishing pastes.**
- Polishing imparts a shine or lustrous surface.
- Polishing process can affect many aspects of the final restoration including **surface staining, plaque accumulation and wear characteristics of composite.**

- Polishing strips are used to contour and polish surface and margin



Criteria for finished composite resin restorations

- Margins should be intact with no voids or deficiencies
- Margins should be flush with the surface
- Anatomy restored
- Surface should be smooth and shiny
- Occlusion checked

Following failures are commonly seen in composite restorations with time

Composite restorations may show failure because of

- Incomplete removal of carious lesion
- Incomplete etching or incomplete removal of residual acid from tooth surface
- Excess or deficient application of bonding agent
- Lack of moisture control
- Contamination of composite with finger/saliva
- Following bulk placement technique during polymerization of composite
- Improper polymerization method
- Incomplete finishing and polishing of composites
- Inadequate occlusion of restored tooth.



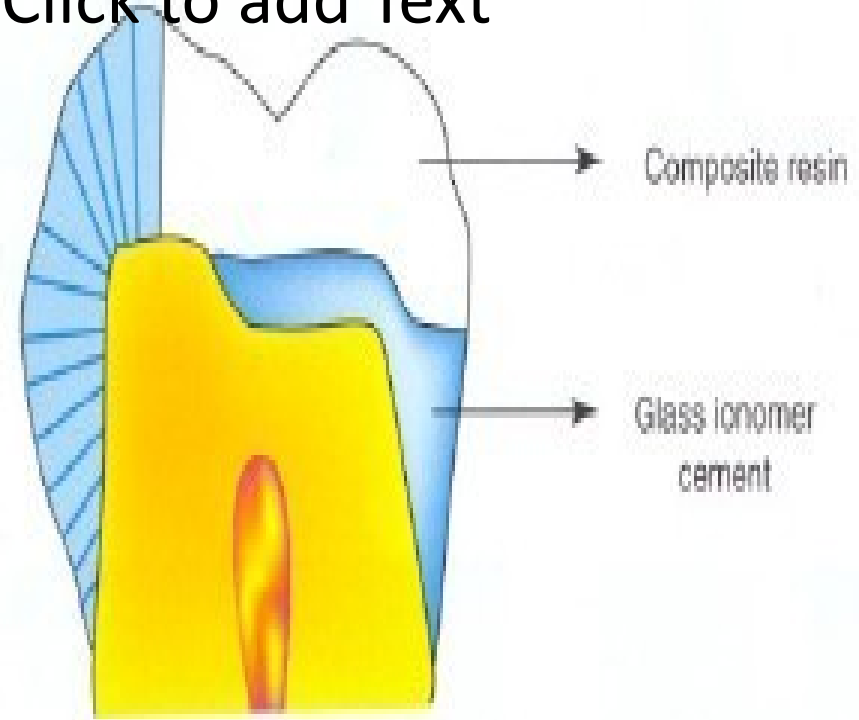
Following failures are commonly seen in composite restorations with time

- Discoloration, especially at the margins ([REDACTED])
- Fracture of margins
- Secondary caries
- Postoperative sensitivity
- Gross fracture of restoration
- Loss of contact after a period of time
- Accumulation of plaque around the restorations.

Bilayered/sandwich technique

- GIC - replace carious dentin prior to attachment of composite resins to etched enamel
- decrease microleakage compared to simple composite restoration.
- clinically employed while restoring large Class III, Class IV, Class V, Class I and Class II cavities with direct composite resin

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➤ Reference

- Text book of operative dentistry 3rd edition
- Art and science of operative dentistry 7th edition